Alphabets, Strings, (Formal) Languages & Decision Problems

What does it mean to compute?

Alphabet $\Sigma = \{0,1\}$, $\Sigma = \{a,b\}$, $\Sigma = \{\alpha, \beta, \gamma, \delta\}$

convention to name set with capital greek letters.

$\Gamma = \{0, 1, a, b, ☺\}$

$B \longrightarrow$ blank symbol

An alphabet is a finite non-empty set of unit length symbols.

e.g., $\Sigma = \Sigma' = \{0,1\}$

set that contains all strings of length 2 for alphabet Σ. $\leftarrow \Sigma^2 = \Sigma \cdot \Sigma = \{00, 01, 10, 11\}$

↳ concatenation operation allows us to join symbols to get longer strings

$\Sigma^3 = \Sigma \cdot \Sigma \cdot \Sigma = \{000, 001, 010, 011, 100, 101, 110, 111\}$

↳ set containing all strings of length three for alphabet Σ.

Note: $000 \neq 00 \neq 0$ These are strings not numbers

Also, concatenation operation can be overloaded. Can apply to strings & sets.

1. For some strings w_1 & w_2 we get a string $w_3 = w_1 \cdot w_2$.
   e.g., $w_1 = 01$, $w_2 = 10$ $\longrightarrow$ $w_3 = w_1 \cdot w_2 = 0110$

2. $\Sigma \cdot \Gamma$ gives a set that contains all unique strings obtained by concatenating elements from Σ with elements from Γ.
   (order matters)

e.g. $\Sigma = \{0,1\}$ & $\Gamma = \{a,b\} \longrightarrow \Sigma \cdot \Gamma = \Delta = \{0a, 0b, 1a, 1b\}$

We also have a special symbol for a string of length zero.
λ is called the empty string & is defined to have length zero.

So, we can also define $\Sigma^0 = \{\lambda\}$

↳ Set that contains strings of length zero.

Note: $\Sigma^0 = \{\lambda\} \neq \phi$

Set that contains the empty string as an element $\neq$ The empty set has no elements.

We can now generate all possible strings from an alphabet Σ via Σ^*. $\rightarrow$ Kleene Closure

Kleene Closure $\Sigma^* = \bigcup_{i=0}^{\infty} \Sigma^i = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \ldots$

e.g., for $\Sigma = \{0,1\}$, Σ^* contains all binary strings

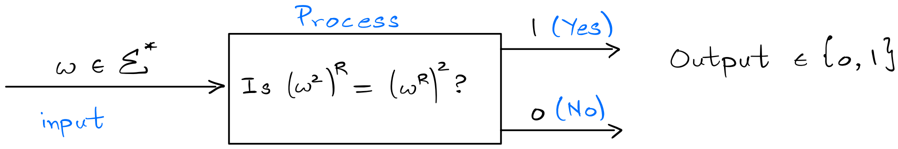
Let $w \in \Sigma^*$, where $\Sigma = \{0,1\}$ $\longrightarrow$ so, w is any possible binary string.

$w^R \longrightarrow$ Reversal operation

e.g., $w = 01101 \longrightarrow w^R = 10110$

For a $w \in \Sigma^*$, is $(w^2)^R$ always equal to $(w^R)^2$?

The process we use to identify which w satisfy the above criterion can be thought of as a computation.



So, we can model computation as solving a decision problem.

It is a mapping from Σ^* to $\{0,1\}$.

The study of computation then is to study this mapping & its properties under various restrictions.

Every computation identifies a language, i.e., the set of strings for which the output is 1.

Language Any subset of Σ^* is a language.